Problem Set 1

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Applied Stats/Quant Methods 1

Due: October 9, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 9, 2025. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
    80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school.

```
#calculate sample mean
#mean is the sum of the elements in vector y divided by the number of
    elements in y.

```

```r
count_y <- length(y)
calc_mean_y <- sum(y)/count_y
# verify with the function
func_mean_y <- mean(y)

#calculate stddev
#evaluate the vector of demeaned values demeaned_y_i = func_mean_y - y_i
y_minus <- y - mean(y)
y_minus_sqr <- y_minus^2

#sum the sqr of the vector elements
demeaned_sum_sqr <- sum(y_minus_sqr)

# calculate variance
# sum of the demeaned values^2 divided by the sample size-1
# n-1 also called degrees of freedom
variance <-demeaned_sum_sqr/(length(y)-1)

#calculate stddev
standard_deviation <- sqrt(variance)

# verify with the function
func_stddev <- sd(y)
#YAY, they match.

#calculate stderr of the sample -- divide stddev by sqrt of sample size
standard_error <- standard_deviation/ sqrt(length(y))

#check with ggplot function
func_mean_se <- (mean_se(y, mult = 1))

# egads, this returns a dataframe.  now i need to figure out how to use that.
# and need absolute value of ....
#(thanks, stack overflow)
func_stderr <- abs(func_mean_se[[2]] - func_mean_se[[1]])

## ----------

#Find the 90% confidence interval for the average student IQ in the school.

z90 <-qt((1 - .90)/2, df= (count_y -1), lower.tail = FALSE)
lower_90 <- func_mean_y - ( z90 * func_stddev/ sqrt(count_y))
upper_90 <- func_mean_y + ( z90 * func_stddev/ sqrt(count_y))
confint90 <- c(lower_90, upper_90)
```

The 90% confidence interval for the average student IQ in our sample is:

```
[94, 104]
```

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

   Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

```r
# ?t.test
# H0 ---  mean_y = mu = 100
# Halpha --- mean_y is either greater or less than mu=100
# two sided t-test
# not terribly useful here... but good practice
mu0 <- 100

# T_stat <- (mean_y  - 100)/ ( std_dev/ sqrt(n) ), degrees of freedom = (length(y) -1)
t_stat <- (calc_mean_y  - mu0)/(standard_error)
#print(t_stat )
#?pt
p_value <-  2*pt(abs(t_stat), df = (length(y) -1), lower.tail = FALSE) # that random 2 cuz two-sided test.  (slide 25)
# print(p_value)

# p_value =< alpha ===> does not support H0. per Week2Slide20

# verify with internal functions
# alternative hypothesis (Halpha) is that the true mean or population parameter is not equal to hypothesized value 100.
two_sided_hypothesis_test <- t.test(y, alternative = "two.sided", mu=100, conf.level = 0.95)
# print(two_sided_hypothesis_test)

##  ----------

# H0 ---  mean_y =< mu = 100
# Halpha ------- mean_y > mu = 100
# one sided t-test
mu0 <- 100

# T_stat <- (mean_y  - 100)/ ( std_dev/ sqrt(n) ), degrees of freedom = (n-1)
t_stat <- (calc_mean_y  - mu0)/(standard_error)
# print(t_stat)

#?pt
one_sided_p_value <-  pt(t_stat, df = (length(y) -1), lower.tail = FALSE)  #  one-sided test.  (slide 25)
# print(one_sided_p_value)

# p_value > alpha = 0.05 ===> supports H0.  sample mean(IQ) =< than population.
# alternative hypothesis (Halpha) was that the true mean of the sample is
```

```
    greater than a hypothesized value 100.
# p_value does not support Halpha

# verify with internal functions
one_sided_hypothesis_test <- t.test(y, alternative = "greater", mu=100,
   conf.level = 0.95)
# print(one_sided_hypothesis_test)
```

A one sided t-test returns a p-value of 0.72 which supports the null hypothesis that the sample mean is less than or equal to 100. Sad school counselor :(.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

`State`	*50 states in US*
`Y`	*per capita expenditure on shelters/housing assistance in state*
`X1`	*per capita personal income in state*
`X2`	*Number of residents per 100,000 that are "financially insecure" in state*
`X3`	*Number of people per thousand residing in urban areas in state*
`Region`	*1=Northeast, 2= North Central, 3= South, 4=West*

Explore the `expenditure` data set and import data into `R`.

```r

expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2025/main/datasets/expenditure.txt", header=T)

# explore the data
# print(expenditure)
str(expenditure)
typeof(expenditure)
dim(expenditure)
head(expenditure)
```

- Please plot the relationships among *Y*, *X1*, *X2*, and *X3*? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```r
# plot(expenditure)

pdf("./problemSets/PS01/my_answers/Q2_Y_X1_X2_X3.pdf")
pairs(expenditure[,2:5], main = "examine relationships between Y, X1, X2, X3", col = expenditure$Region)
dev.off()


regression1 <- lm(X3~X1, data=expenditure)
# now save that output to a file that you can read in later to your answers
# make it easier for when we need to do this again, let's create a function
output_stargazer <- function(outputFile, ...) {
  output <- capture.output(stargazer(...))
```

```
  cat(paste(output, collapse = "\n"), "\n", file=outputFile, append=TRUE)
}

output_stargazer("./problemSets/PS01/my_answers/regression_output_X3_X1.
    tex", regression1)

y <- expenditure$Y
x1 <- expenditure$X1
x2 <- expenditure$X2
x3 <- expenditure$X3

pdf("./problemSets/PS01/my_answers/Q2_prettier_Y_X1_X2_X3.pdf")

plot(x1, y, main="Per capita income vs. \n Expenditure on shelters by
    state", xlab="Income", ylab="Shelter expenditure", col="blue", cex=1)

plot(x2, y, main="Number of residents per 100k that are insecure vs. \n
    Per capita expenditure on shelters by state",
     xlab="Number of residents per 100k that are insecure", ylab="Shelter
     expenditure", col="blue", cex=1)

plot(x3, y, main="Number of people per 1k residing in urban areas vs. \n
    Per capita expenditure on shelters by state",
     xlab="Number of people per 1k residing in urban areas", ylab="
    Shelter expenditure", col="blue", cex=1)
dev.off()

pdf("./problemSets/PS01/my_answers/Q2_spending_by_region_colored.pdf")
ggplot(data = expenditure, aes(x=X1,y =Y, color=Region)) + geom_point() +
  labs(
    x = "Per capita personal income in state",
    y = "Per capita expenditure on shelters/housing assistance in state",
    colour = "regions",
    title = "Per capita personal income in state vs.\nPer capita
    expenditure on shelters/housing assistance in state ",
    subtitle = "Colored by region"
  )
dev.off()
```

Figure 1: relationships among Y, $X1$, $X2$, and $X1$.

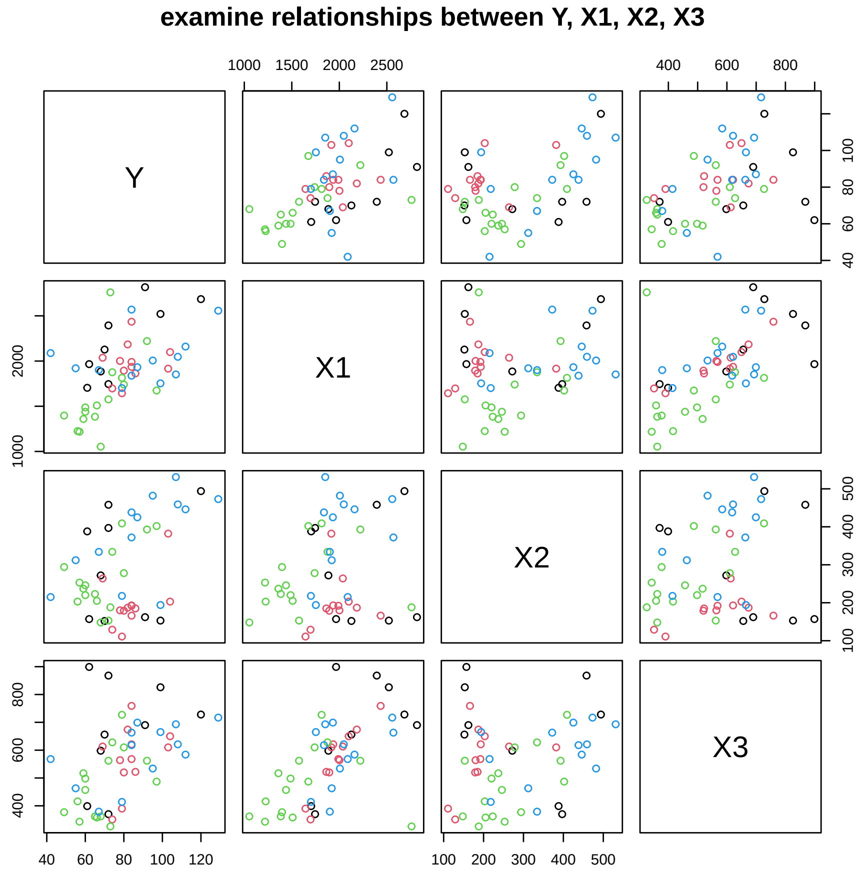


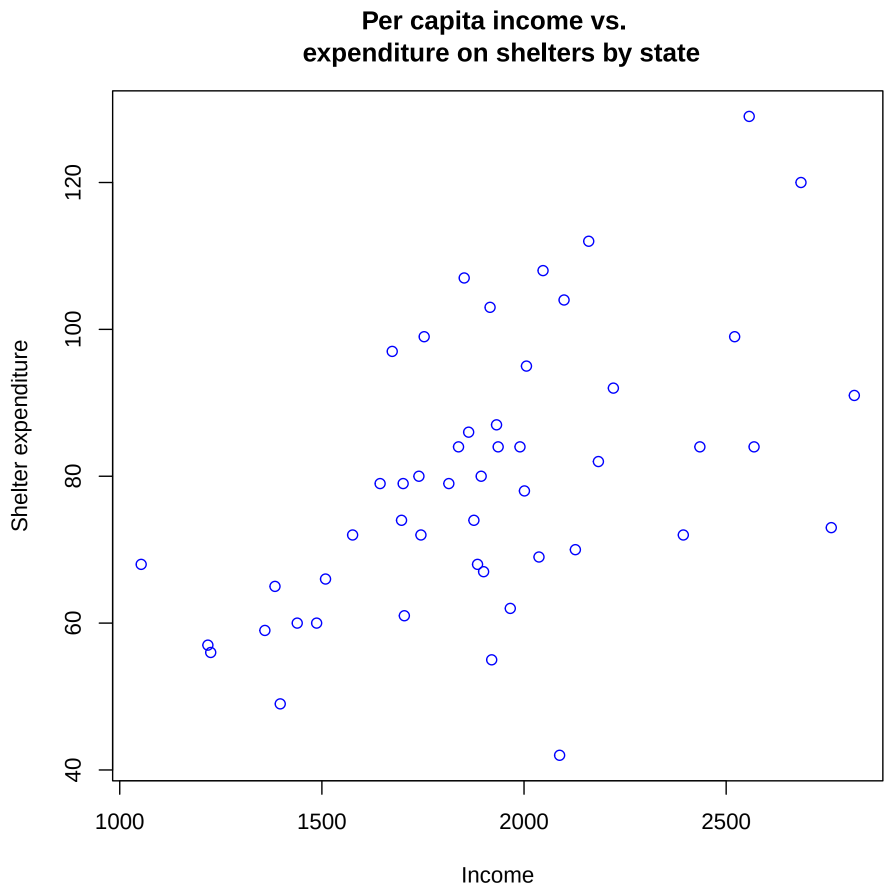
Table 1:

	Dependent variable:
	X3
X1	0.216***
	(0.042)
Constant	149.428*
	(82.040)
Observations	50
R^2	0.354
Adjusted R^2	0.341
Residual Std. Error	117.750 (df = 48)
F Statistic	26.341*** (df = 1; 48)
Note:	*p<0.1; **p<0.05; ***p<0.01

In the pairwise compairisons of the variables Y, X1, X2, and X3 in addition to the specific Y vs X1, X2, and X3 also plotted below, there is an simple positive linear correlation with a slope of 0.21 in the graph of X3 vs X1 "Number of people per thousand residing in urban areas in state vs per capita personal income in state". Though, *Table 1* shows a small R^2 value leave a large (>0.5) unexplained variance.

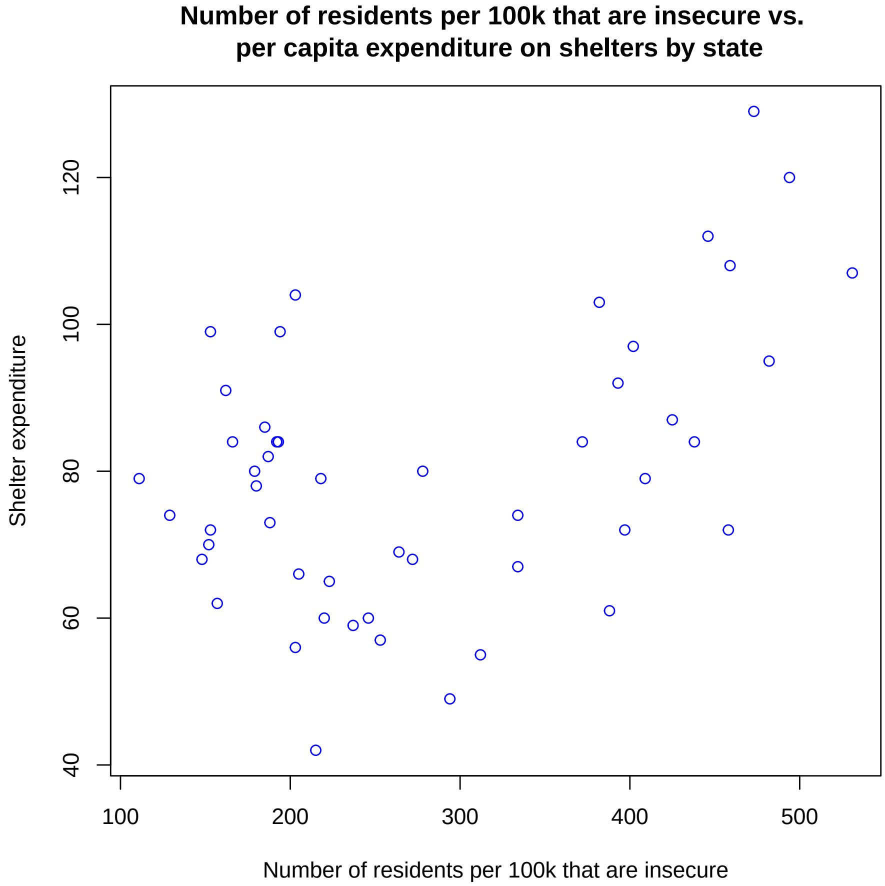
Additionally, there are interesting features in the graph of X2 vs X1 "Number of residents per 100,000 that are *financially insecure* in state vs. per capita personal income in state" with two ranges of X2 where X1 varies across the range of "per capita personal income in the state"

Figure 2: relationships between Y and $X1$.



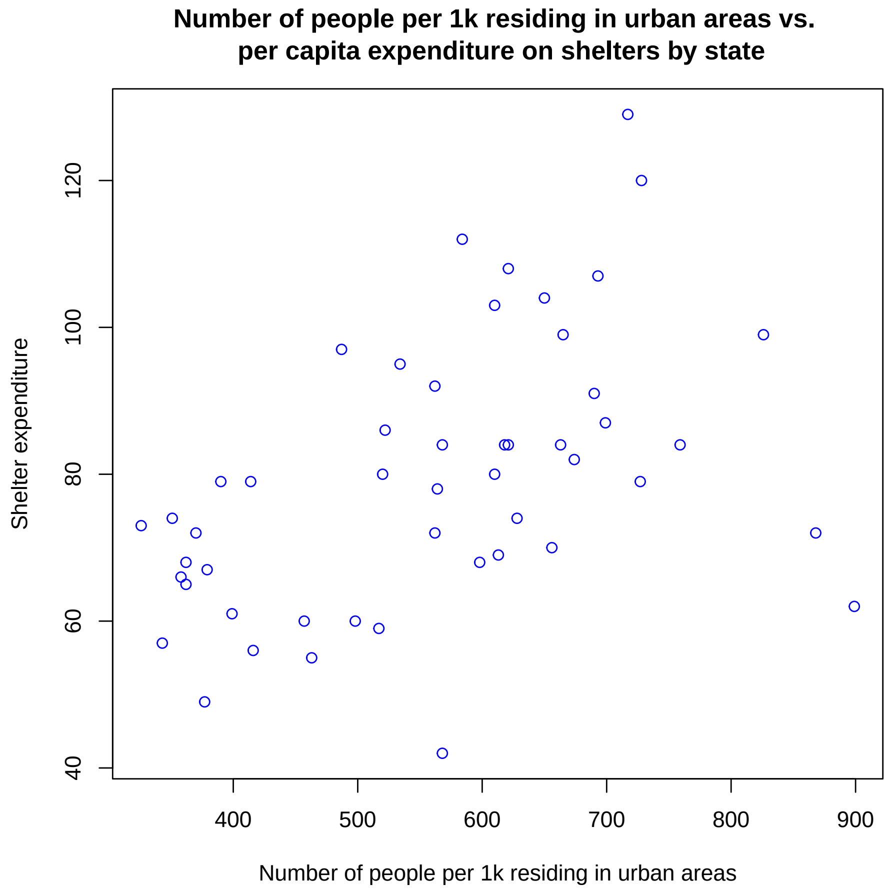
There is a positive correlation between shelter expenditure and per capita income.

Figure 3: relationships between Y and $X2$.



There appears to be a u shaped correlation where the shelter expenditure first decreases as the proportion of residents who are insecure increases and then that trend reverses and begins increasing again after the x value of approximately 300/100k .

Figure 4: relationships between Y and $X3$.



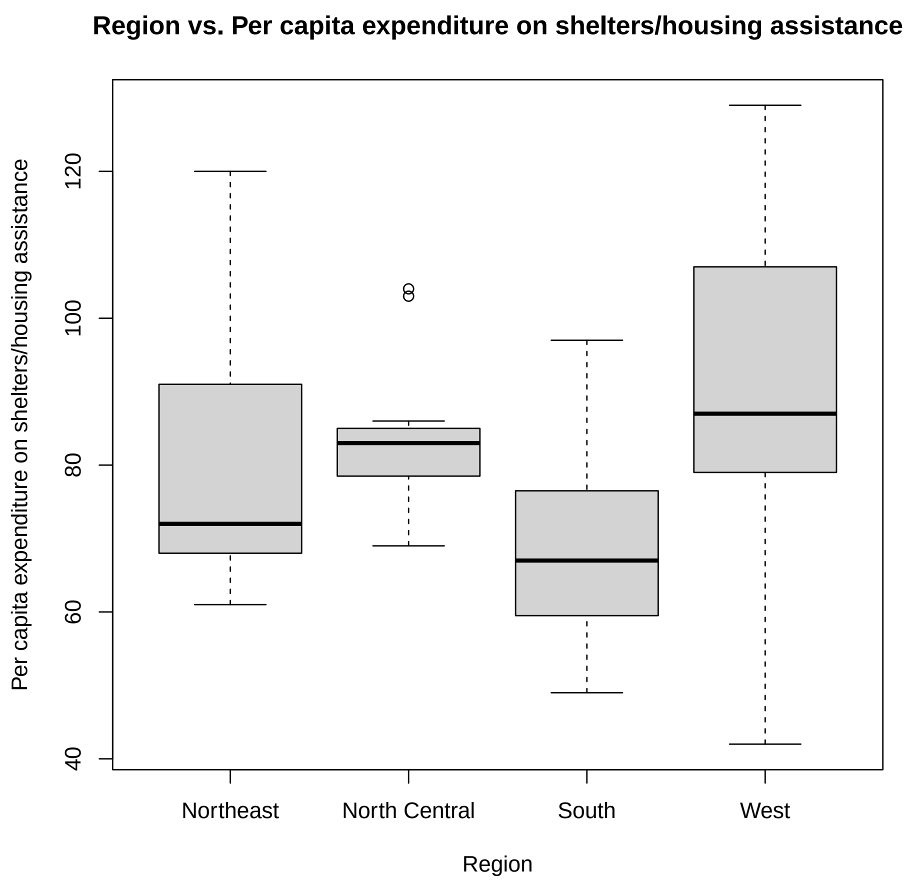
There appears to be a linear positive correlation between number of people / one thousand living in urban areas and per capita expenditure on shelters with some strange outliers in very densely populated states.

- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```
pdf("./problemSets/PS01/my_answers/Q2_average_spending_by_region.pdf")

# bah, can't get the x axis grouping to number correctly.
# ggplot(data = expenditure, aes(group=Region,y =Y)) + geom_boxplot() +
#   labs(
#     x = "Region",
#     y = "Per capita expenditure on shelters/housing assistance",
#     title = "Region vs. Per capita expenditure on shelters/housing assistance",
#     subtitle = "Box Plot showing average per region"
```

```
#    )

region_names <- c("Northeast", "North Central", "South", "West")
boxplot(expenditure$Y ~ expenditure$Region,
        names=region_names,
  main = "Region vs. Per capita expenditure on shelters/housing
    assistance",
  xlab = "Region",
  ylab = "Per capita expenditure on shelters/housing assistance")

dev.off()
```

Figure 5: relationships between Y and *Region*.

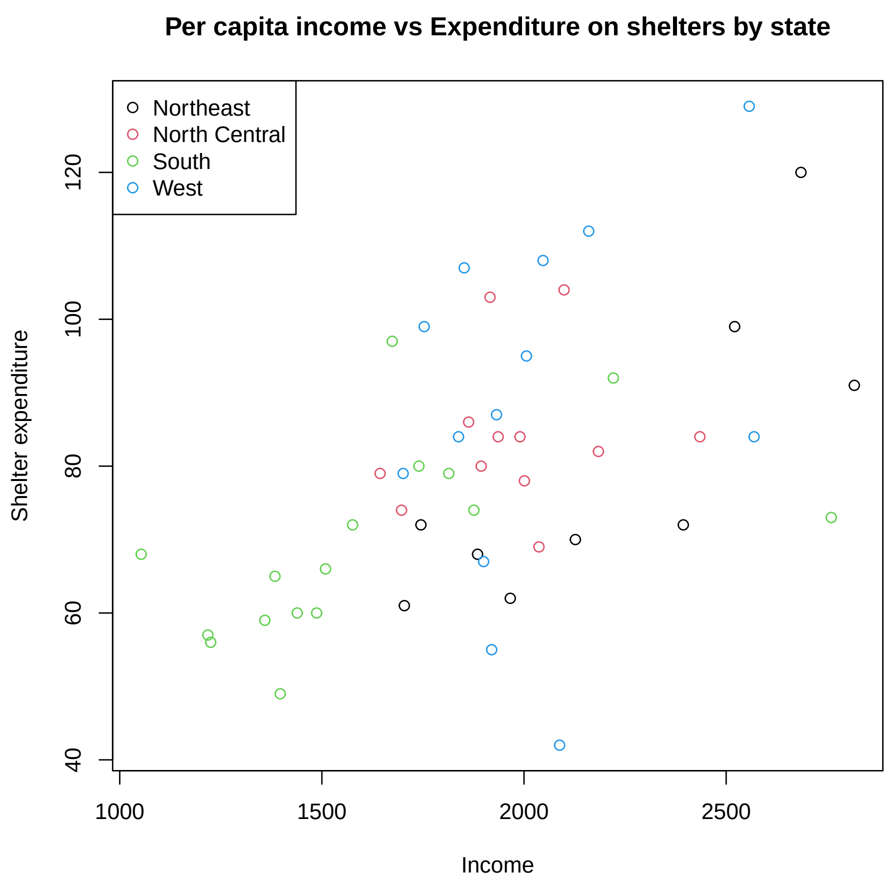


On average, the West has the greatest per capita spending on shelter/housing assistance.

- Please plot the relationship between Y and *X1*? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```
pdf("./problemSets/PS01/my_answers/Q2_scatter_spending_by_region_colored.pdf")
plot(expenditure$X1, expenditure$Y, main="Per capita income vs Expenditure on
   shelters by state", xlab="Income",
     ylab="Shelter expenditure", col= expenditure$Region, cex=1)

legend("topleft",
       legend=region_names,
       col=1:length(region_names),
       pch=1)

dev.off()
```

Figure 6: relationships between *Y* and *X1* with *Region* by color.



There is a positive correlation between per capita shelter expenditure and per capita income with large variance and the South region making a particularly poor showing.

Table 2:

	Dependent variable:
	X1
Y	11.502***
	(2.644)
Constant	996.992***
	(215.844)
Observations	50
R^2	0.283
Adjusted R^2	0.268
Residual Std. Error	342.576 (df = 48)
F Statistic	18.920*** (df = 1; 48)
Note:	*p<0.1; **p<0.05; ***p<0.01